

Course Name

Course Code: PH113
Physics

Teacher's Name:

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Department of Physics

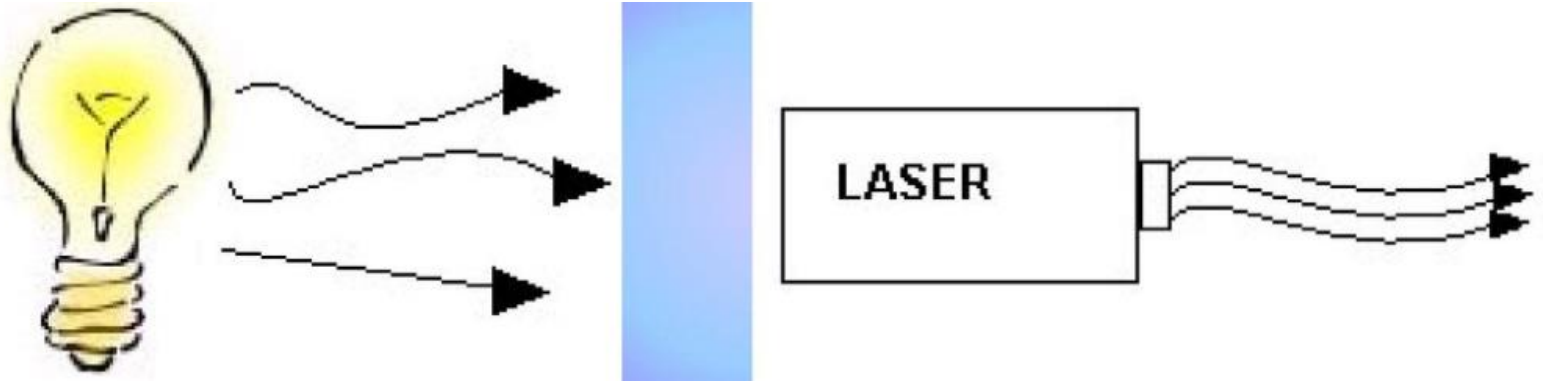
Module-5

Introduction to Laser

Laser

A laser is a device that emits light through a process of optical amplification based on the stimulated emission of electromagnetic radiation. The word "laser" is an acronym for "Light Amplification by Stimulated Emission of Radiation".

Incandescent vs Laser Light

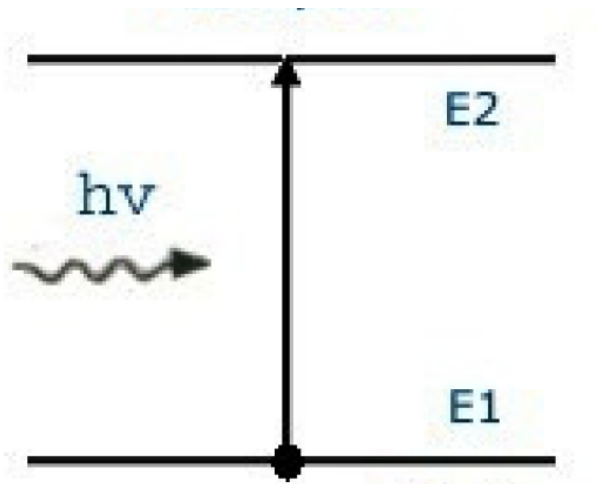


- Many Wavelength
- Multidirectional
- Incoherent

- Monochromatic
- Directional
- Coherent

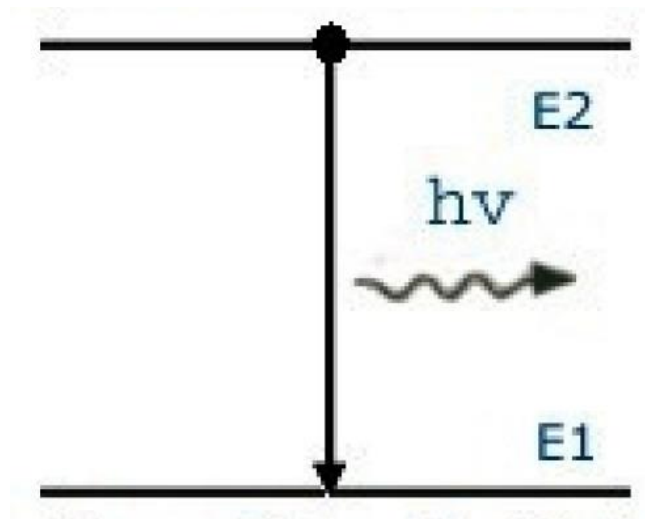
Stimulated Absorption

An atom in the lower energy level or ground state energy level E_1 absorbs the incident photon radiation of energy $h\nu$ and goes to the higher energy level or excited level E_2 . This process is called absorption.



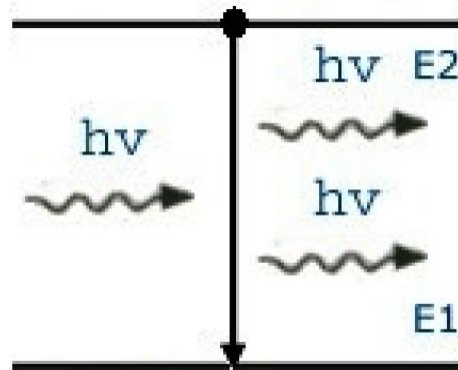
Spontaneous Emission

The atom in the excited state returns to the ground state by emitting a photon of energy $E=(E_2-E_1)=h\nu$ spontaneously without any external triggering as shown in the figure. This process is known as spontaneous emission. Such an emission is random and is independent of incident radiation.



Stimulated Emission

The atom in the excited state can also return to the ground state by external triggering or inducement of photon thereby emitting a photon of energy equal to the energy of the incident photon, known as stimulated emission. Thus results in two photons of same energy, phase difference and of same directionality.

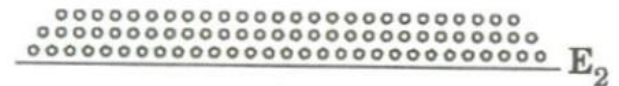
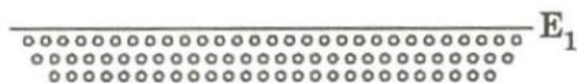


Stimulated vs Spontaneous Emission

Stimulated Emission	Spontaneous emission
An atom in the excited state is induced to return to the ground state , thereby resulting in two photons of same frequency and energy is called Stimulated emission	The atom in the excited state returns to the ground state thereby emitting a photon, without any external inducement is called Spontaneous emission.
The emitted photons move in the same direction and is highly directional	The emitted photons move in all directions and are random
The radiation is highly intense, monochromatic and coherent	The radiation is less intense and is incoherent
The photons are in phase, there is a constant phase difference	The photons are not in phase (i.e.) there is no phase relationship between them .

Population Inversion

Population Inversion creates a situation in which the number of atoms in higher energy state is more than that in the lower energy state. Usually at thermal equilibrium, the number of atoms N_2 i.e., the population of atoms at higher energy state is much lesser than the population of the atoms at lower energy state N_1 that is $N_1 > N_2$. The Phenomenon of making $N_2 > N_1$ is known as Population Inversion.



Conditions for Population Inversion

1. There must be at least two energy levels $E_2 > E_1$.
2. There must be a source to supply the energy to the medium.
3. The atoms must be continuously raised to the excited state.

Meta Stable States

An atom can be excited to a higher level by supplying energy to it. Normally, excited atoms have short lifetimes and release their energy in a matter of nano seconds (10^{-9}) through spontaneous emission. It means atoms do not stay long to be stimulated. As a result, they undergo spontaneous emission and rapidly return to the ground level; thereby population inversion could not be established. In order to do so, the excited atoms are required to 'wait' at the upper energy level till a large number of atoms accumulate at that level. In other words, it is necessary that excited state have a longer lifetime.

Meta Stable States

A Meta stable state is such a state. **Metastable state** is an excited state of an atom or other system with a longer lifetime than the other excited states. Metastable can be readily obtained in a crystal system containing impurity atoms. These levels lie in the forbidden gap of the host crystal. There could be no population inversion and hence no laser action, if metastable states don't exist.

Pumping

The process of raising large number of atoms from lower energy to a higher energy level is called pumping.

Types of pumping:

- Optical pumping: which uses strong light source for excitation.
- Electrical pumping: which uses electron impact for excitation.
- Chemical pumping: which uses chemical reactions for excitation.
- Direct pumping: which uses direct conversion of electric energy into light energy.

Optical Pumping

Optical pumping, in physics, the use of light energy to raise the atoms of a system from one energy level to another. A system may consist of atoms having a random orientation of their individual magnetic fields. When optically pumped, the atoms will undergo a realignment of individual magnetic fields with respect to the direction of the light beam; that is, there will be a rearrangement of magnetic energy levels; and, in some cases, the bulk of the atoms will be oriented to generate a magnetic field throughout the system.

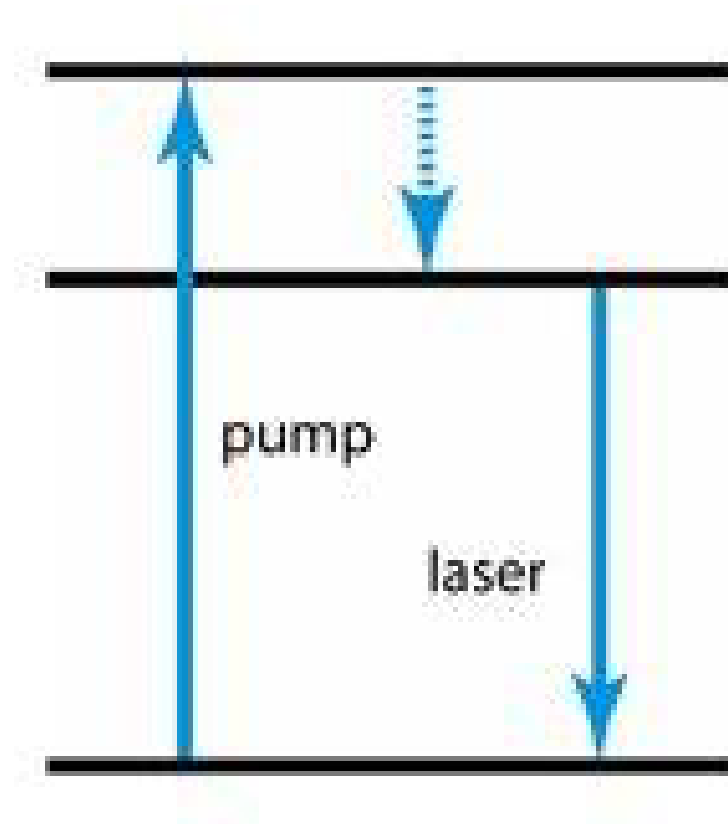
Optical Pumping

Optical pumping is a process in which light is used to raise (or "pump") electrons from a lower energy level in an atom or molecule to a higher one. It is commonly used in laser construction **to pump the active laser medium so as to achieve population inversion.**

Optical Pumping

Inversion by optical pumping becomes possible when using a three-level system. Pump light with a shorter wavelength (higher photon energy) can transfer atoms from the ground state to the highest level. From there, spontaneous emission or a non-radiative process (e.g., involving phonons in a laser crystal) transfers atoms to an intermediate level, called the upper laser level. From that level down to the ground state, the laser transition with stimulated emission can occur. With sufficiently high pump intensity, population inversion for the laser transition can be reached as stimulated emission by the pump radiation is prevented by the transfer to the intermediate level.

Optical Pumping



Einstein's A & B Coefficients

We know that, when light is absorbed by the atoms or molecules, then it goes from the lower energy level (E_1) to the higher energy level (E_2) and during the transition from higher energy level (E_2) to lower energy level (E_1) the light is emitted from the atoms or molecules. Let us consider an atom exposed to light photons of energy, three distinct processes take place.

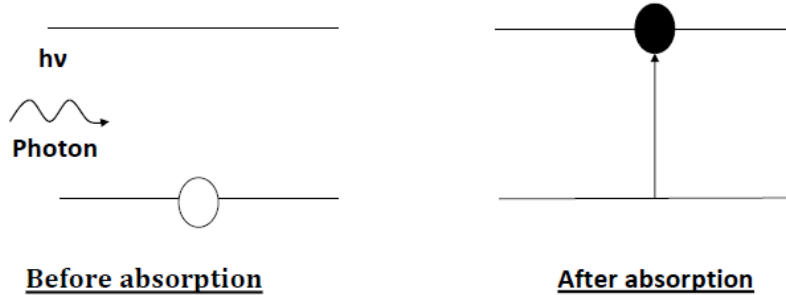
- Absorption
- Spontaneous emission
- Stimulated Emission

Einstein's A & B Coefficients

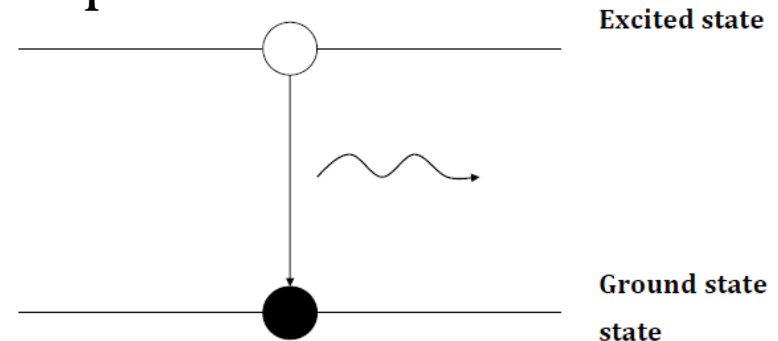
The Einstein A coefficients are related to the rate of spontaneous emission of light, and the Einstein B coefficients are related to the absorption and stimulated emission of light.

Einstein's A & B Coefficients

Absorption



Spontaneous Emission



Rate of absorption is given by,

$$R_{AB} \propto \rho N_1$$

$$R_{AB} = B_{12} \rho N_1$$

Rate of spontaneous emission is given by,

$$R_{SP} \propto N_2$$

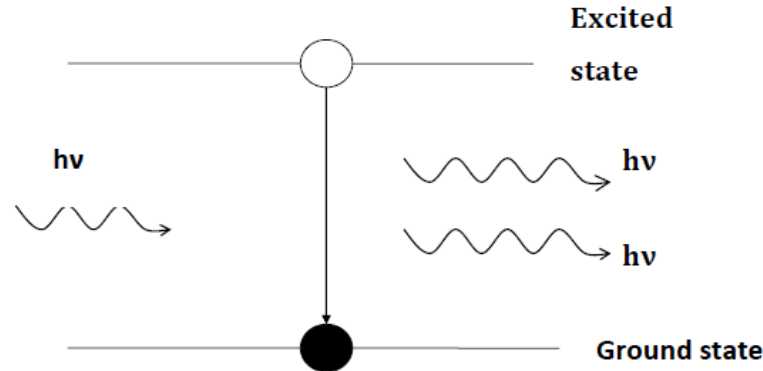
$$R_{SP} = A_{21} N_2$$

where

- N_1 and N_2 are the number of atoms in the lower and higher energy states respectively and
- ρ is Energy density of incident radiation,
- B_{12} - is a constant which gives the probability of absorption transition per unit time.
- A_{21} - is a constant which gives the probability of spontaneous emission transitions per unit time.

Einstein's A & B Coefficients

Stimulated emission



Rate of stimulated emission is given by,

$$R_{ST} \propto \rho N_2$$

$$R_{ST} = B_{21} \rho N_2$$

where

- N_2 are the number of atoms in higher energy states
- ρ is Energy density of incident radiation,
- B_{21} - is a constant which gives the probability of stimulated emission transitions per unit time.

Einstein's A & B Coefficients

Under thermal equilibrium, the mean population N_1 and N_2 in the lower and upper energy levels respectively remains constant.

This condition requires that the number of transition from E_2 to E_1 must be equal to number of transition from E_1 to E_2

Thus, The rate of absorption = The rate of emission

$$R_{AB} = R_{SP} + R_{ST}$$

$$B_{12} \rho N_1 = A_{21} N_2 + B_{21} \rho N_2$$

$$\rho[B_{12} N_1 - B_{21} N_2] = A_{21} N_2$$

Einstein's A & B Coefficients

$$\rho = \frac{A_{21}N_2}{B_{12}N_1 - B_{21}N_2}$$
$$\rho = \frac{A_{21}}{B_{21}} \cdot \frac{1}{\left(\frac{B_{12}}{B_{21}} \frac{N_1}{N_2} - 1\right)} \text{ --- } (1)$$

From Boltzmann distribution law, the relative population

Where,

$$\frac{N_1}{N_2} = e^{(E_2 - E_1)/KT} \text{ --- } (2)$$

K → Boltzmann constant

T → absolute temperature

Einstein's A & B Coefficients

Since $E_2 - E_1 = h\nu$

we have

$$\frac{N_1}{N_2} = e^{(h\nu)/KT} \text{ -----} \rightarrow (3)$$

$$\therefore \rho = \frac{A_{21}}{B_{21}} \cdot \frac{1}{\left(\frac{B_{12}}{B_{21}} e^{h\nu/KT} - 1\right)} \text{ -----} \rightarrow (4)$$

According to Planck's theory of radiation, the energy distribution is given by

$$\rho = \frac{8\pi h\nu^3}{c^3} \cdot \frac{1}{(e^{h\nu/KT} - 1)} \text{ -----} \rightarrow (5)$$

On comparing equations (4) and (5), we can write

$$\frac{A_{21}}{B_{21}} \cdot \frac{1}{\left(\frac{B_{12}}{B_{21}} e^{h\nu/KT} - 1\right)} = \frac{8\pi h\nu^3}{c^3} \cdot \frac{1}{(e^{h\nu/KT} - 1)}$$

Einstein's A & B Coefficients

Where

$$B_{12} = B_{21} = B \text{ -----} \rightarrow (6)$$

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h \gamma^3}{c^3} \text{ -----} \rightarrow (7)$$

$$A = B \left(\frac{8\pi h \gamma^3}{c^3} \right) \text{ -----} \rightarrow (8)$$

Equation (6) and (7) known as Einstein relations and (8) gives the relation between the A and B Einstein's coefficients.

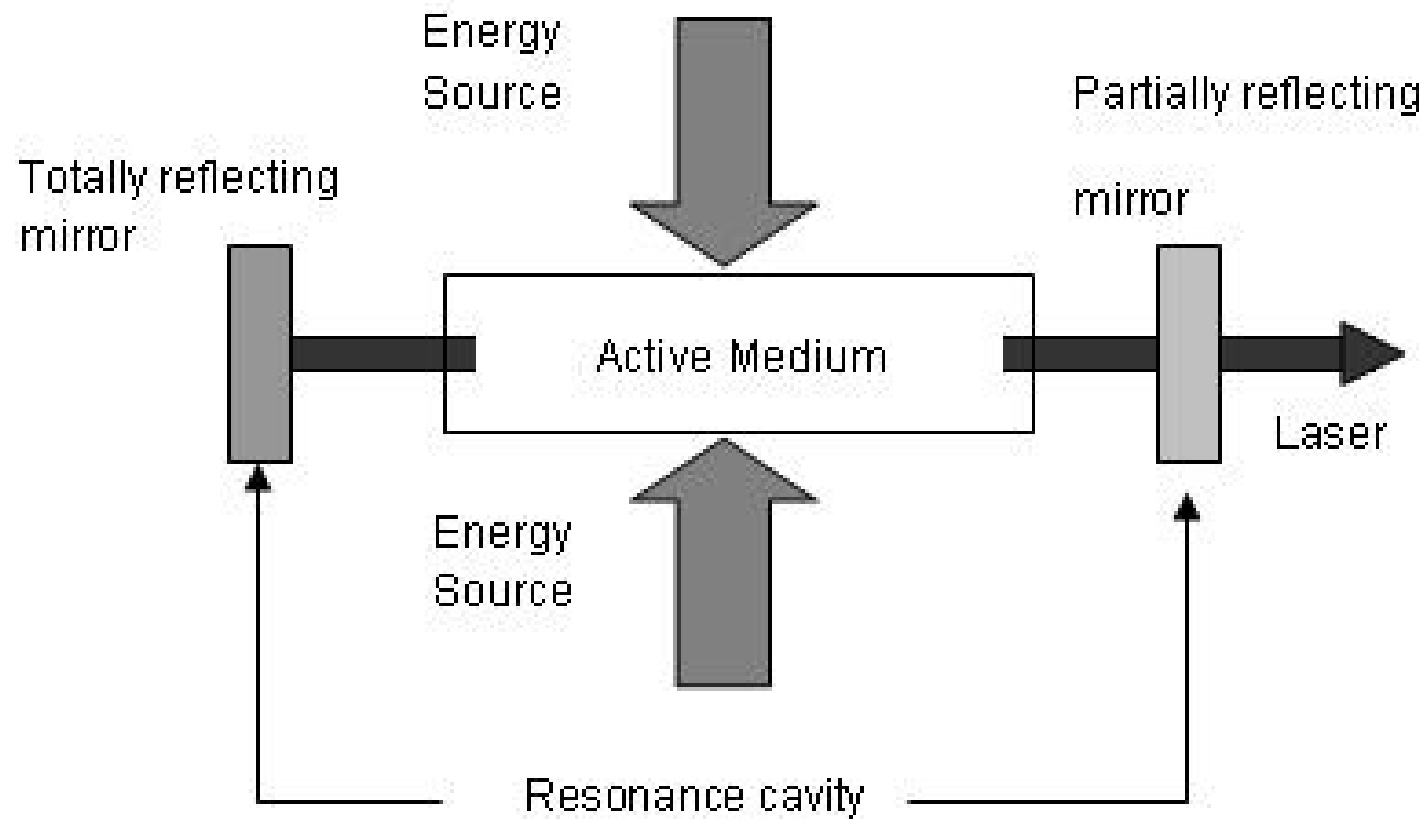
Components of Laser

1. **Active Medium** :It is the material in which the laser action takes place. The active medium may be solid crystals such as ruby or Nd:YAG, liquid dyes, gases like CO₂ or Helium / Neon, or semiconductors such as GaAs. This medium decides the wavelength of laser radiation. Active mediums contain atoms which can produce more stimulated emission than spontaneous emission and cause amplification they are called “*Active Centers*”.

Components of Laser

2. Energy Source (Excitation Mechanism): Energy Source (Excitation mechanisms) pumps the active centers from ground state to excited state to achieve population inversion. The pumping by energy source can be optical, electrical or chemical depending on the active medium.

3. Resonance Cavity: Resonance cavity consists of active medium enclosed between two mirrors one is *highly reflective mirror* (100% reflective) and the other is *partially transmissive mirror* (99% reflective).



Types of Lasers

Based on their gain medium, lasers are classified into five main types:

- Gas Lasers.
- Solid-State Lasers.
- Fiber Lasers.
- Liquid Lasers (Dye Lasers)
- Semiconductor Lasers (Laser Diodes)

Characteristics of Lasers

Laser light has four unique characteristics that differentiate it from ordinary light: these are;

- Coherence
- Directionality
- Monochromatic
- High intensity

Characteristics of Lasers

- **Coherence**: All the photons emitted in laser have the same energy, frequency, or wavelength. Hence, the light waves of laser light have single wavelength or color. Therefore, the wavelengths of the laser light are in phase in space and time. Thus, light generated by laser is highly coherent. Because of this coherence, a large amount of power can be concentrated in a narrow space.
- **Directionality**: in laser, all photons will travel in same direction. Therefore, laser emits light only in one direction. This is called directionality of laser light. The width of a laser beam is extremely narrow. Hence, a laser beam can travel to long distances without spreading.

Characteristics of Lasers

- **Monochromatic:** In laser, all the emitted photons have the same energy, frequency, or wavelength. Hence, the light waves of laser have single wavelength or color. Therefore, laser light covers a very narrow range of frequencies or wavelengths.
- **High Intensity:** In laser, the light spreads in small region of space and in a small wavelength range. Hence, laser light has greater intensity when compared to the ordinary light.

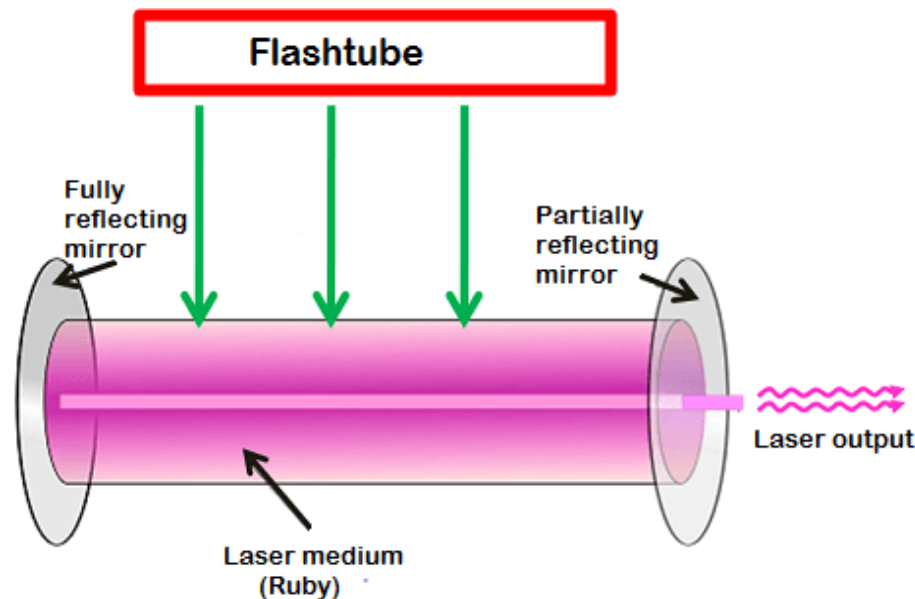
Applications of Laser

There are many applications for laser technology including the following:

- Laser Range Finding.
- Information Processing
- Bar Code Readers.
- Laser Surgery.
- Holographic Imaging.
- Laser Spectroscopy.
- Laser Material Processing.

Ruby Laser

The first working laser was built in 1960 by Maiman, using a ruby crystal and so called the Ruby laser. Ruby belongs to the family of gems consisting of Al_2O_3 with various types of impurities. For example, pink Ruby contains 0.05% Cr atoms. The schematic diagram of ruby laser can be drawn as:



Construction of Ruby Laser

The ruby laser consists of a ruby rod, which is made of chromium doped ruby material. The active material in the Ruby is chromium ion. That is the energy level of chromium Ion takes part in the lasing action. At the opposite ends of this rod there are two silver polished mirrors. One is fully polished and other is partially polished. A spring is attached to the rod with fully polished end for adjustment of wavelength of the laser light. Around the ruby rod a flashlight is kept for the pump input.

The whole assembly is kept in the glass tube. Around the neck of the glass tube the R.F source and switching control is designed in order to switch on and off the flashlight for desired intervals.

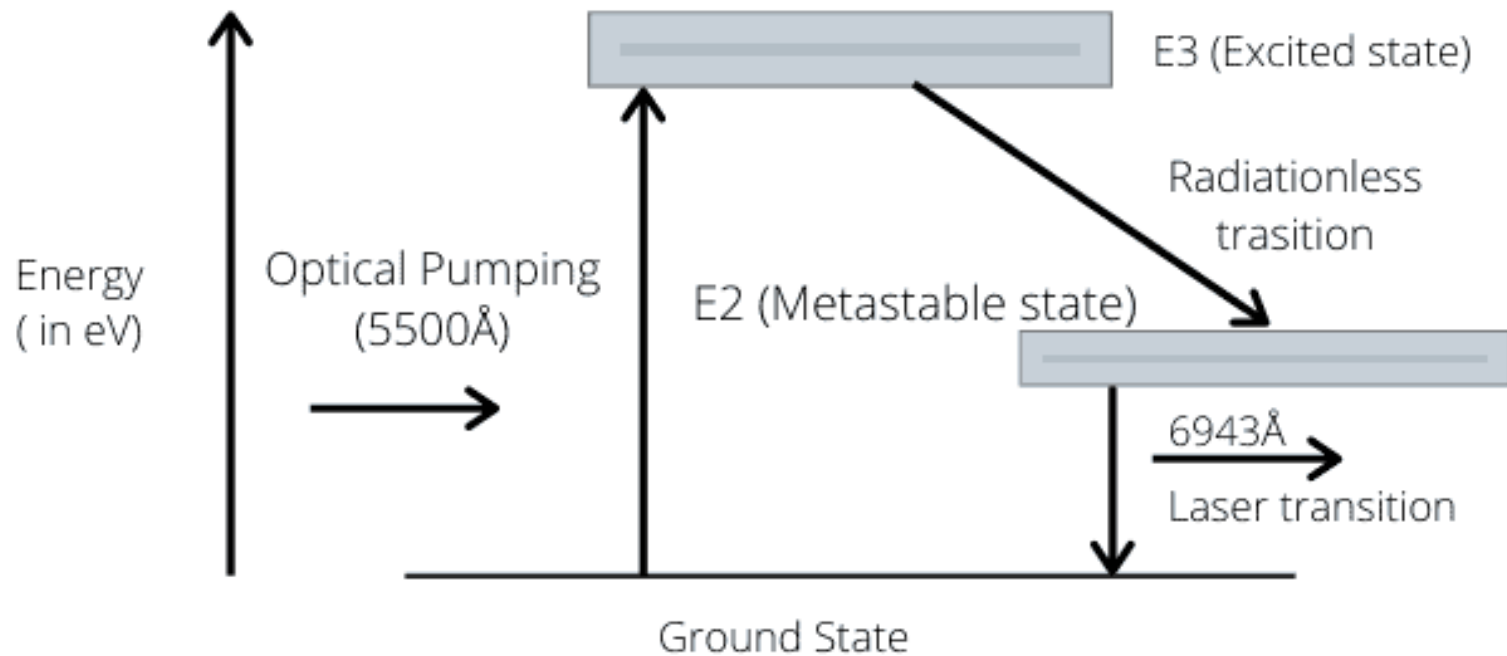
Operation of Ruby Laser

When we switch on the circuit the R.F operates. As a result, the flash of light is obtained around the ruby rod this flash causes the electrons within ruby rod to move from lower energy band towards higher. This flash causes the electrons within ruby rod to move from lower energy band towards higher energy band. The population inversion take place at high energy band and electrons starts back to travel towards the lower energy band. During this movement the electron emits the laser light. This emitted light travels between the two mirrors where cross reflection takes place of this light.

Operation of Ruby Laser

The stimulated laser light now escapes from partially polished mirror in shape of laser beam. The spring attached with the fully polished mirror is used to adjust the wavelength equal to $\lambda/2$ of laser light for optimum laser beam. The switching control of the R.F source is used to switch on and off the flash-light so that excessive heat should not be generated due to very high frequency of the movement of the electron.

Energy Level Diagram of Ruby Laser



Operation of Ruby Laser

The Ruby rod is arranged along the axis of a helical xenon flash tube. In such a manner that the will of the helix encloses the rod. The flash of the tube lasts several milliseconds. The heat produced is removed by liquid nitrogen circulating around the Ruby rod. The remaining part of the energy in the form of blue and green radiation is absorbed by the Ruby. This energy ensures the excitation of chromium Ion from the ground state to the excited energy state for achieving population inversion.

Operation of Ruby Laser

The optical pumping result when incident photons of wavelength $5500\text{\AA}$. Raise the chromium Ion from ground state E_1 to higher excited state E_3 . These iron interact with the crystal lattice and decay to metastable state E_2 by spontaneous emission where they can stay for a longer period ($3 \times 10^{-3}\text{sec}$) producing population inversion. That is, the number of chromium ions in the energy state E_2 (metastable state) is more than the number of chromium ions in excited energy state E_3 . Hence, population inversion is achieve by optical pumping source.

Operation of Ruby Laser

The some of chromium Ion jumps from the metastable state E_2 to the ground state E_1 through spontaneous emission of radiation. During this transition, a photon of wavelength $6943\text{\AA}$ is emitted. This Photon Travels through the Ruby rod in a direction parallel to the axis of the rod and reflects back and forth by the reflecting ends of the optical resonator until it Stimulates an excited chromium ion. Stimulated excited chromium Ion emits a photon exactly in phase with the stimulating photon.

Operation of Ruby Laser

The Stimulated transition of chromium Ion from metastable state E_2 to the ground state E_1 is known as **laser transition**. These in-phase photons stimulate more chromium ions and hence the number of photons emitted increases.

This process repeats and hence the in-phase photons get multiplied. Thus, a strong and coherent laser beam is obtained. Which comes out of the optical resonator through the partially reflecting face.

Advantages of Ruby Laser

- Beam diameter of the ruby laser is comparatively less than CO₂ gas lasers.
- Output power of Ruby laser is not as less as in He-Ne gas lasers.
- Since the ruby is in solid form therefore there is no chance of wasting material of active medium.
- Construction and function of ruby laser is self-explanatory.

Disadvantages of Ruby Laser

- In ruby lasers no significant stimulated emission occurs, until at least half of the ground state electrons have been excited to the Meta stable state.
- Efficiency of ruby laser is comparatively low.
- Optical cavity of ruby laser is short as compared to other lasers, which may be considered a disadvantage.

Applications of Ruby Laser

- Due to low output power they are class-I lasers and so may used as toys for children's.
- It can be used in schools, colleges, universities for science programs.
- It can be used as decoration piece & artistic display.

He-Ne Laser

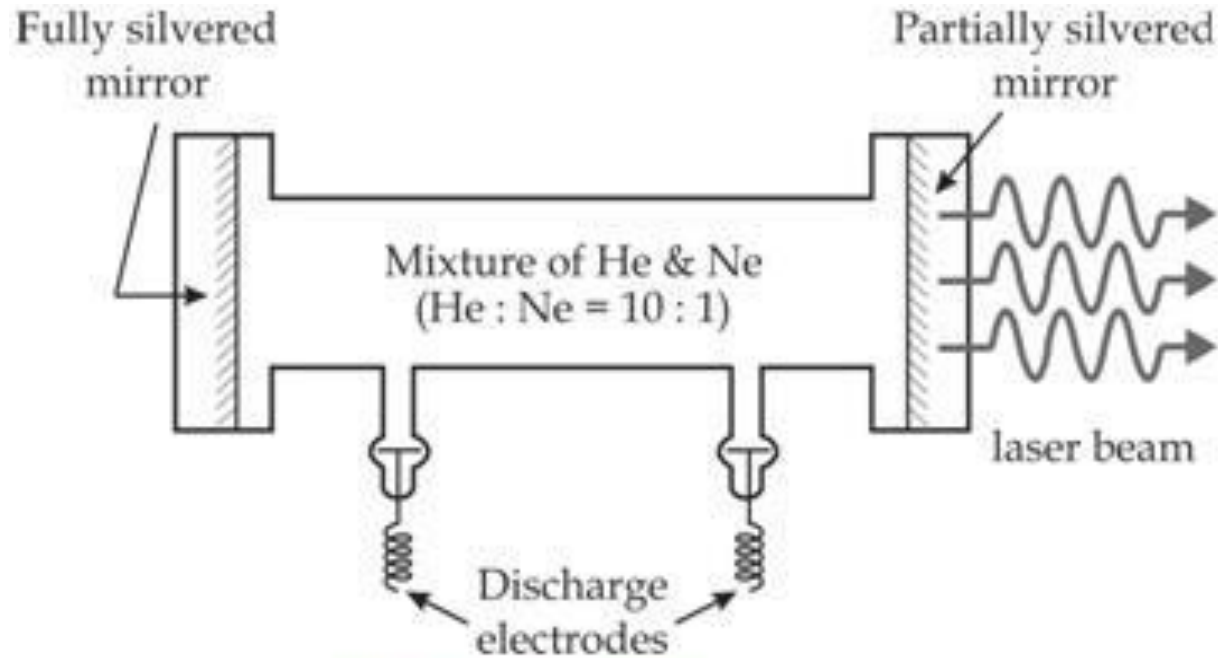
Helium-Neon laser is a type of gas laser in which a mixture of helium and neon gas is used as a gain /active medium. Helium-Neon laser is also known as He-Ne laser.

Construction of He-Ne Laser

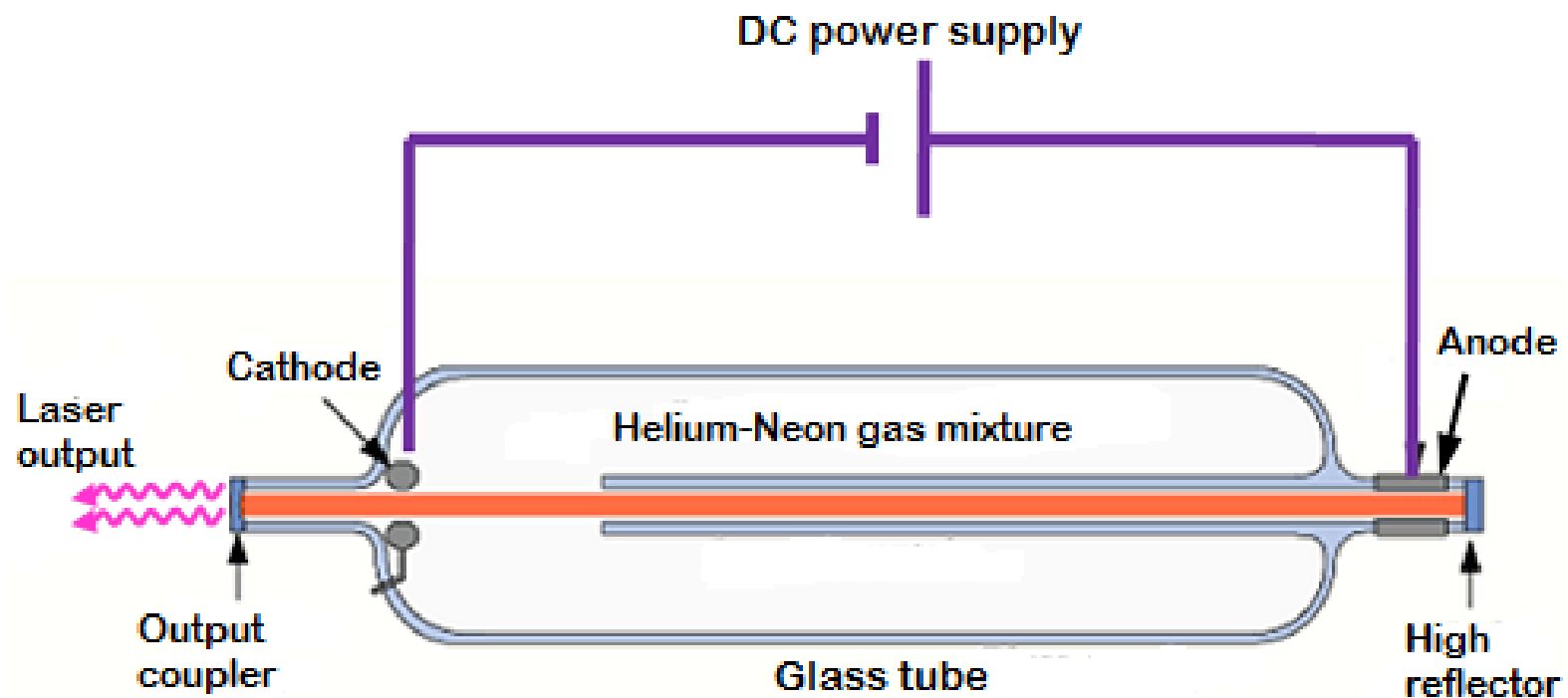
(i) Active medium:

It is a gas laser, which consists of a narrow quartz tube filled with a mixture of helium and neon gases in the ratio 10:1 respectively, at low pressure (~ 0.1 mm of Hg). Ne atoms act as active centres and responsible for the laser action, while He atoms are used to help in the excitation process. The length of the quartz tube is about 50 cm and the diameter is about 1 cm.

Schematic Diagram of He-Ne Laser



He-Ne LASER



Construction of He-Ne Laser

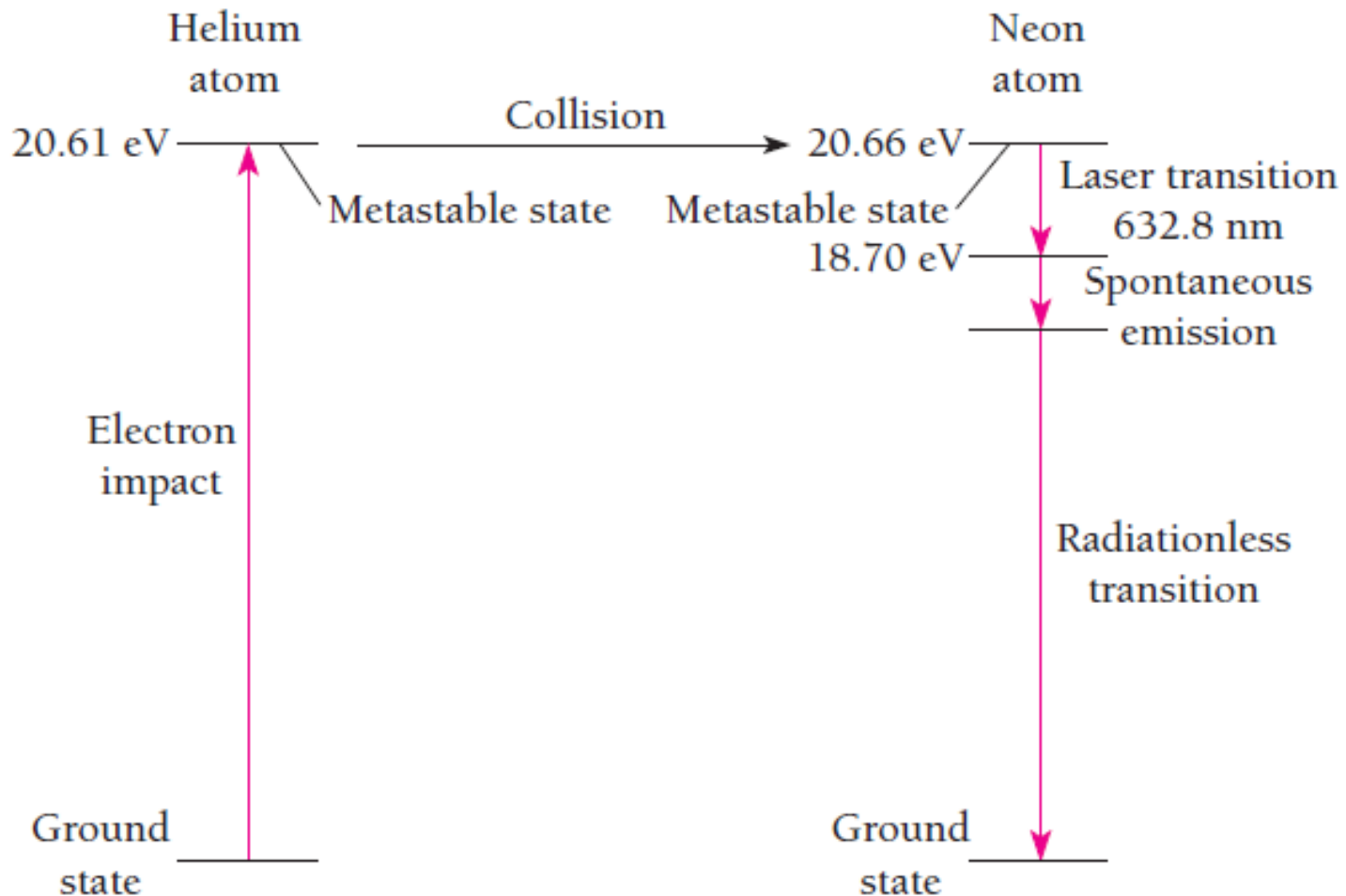
ii) Optical resonator:

To construct the optical resonator cavity, two parallel mirrors are placed at the ends of the quartz tube one of them is partly transparent while the other is fully reflecting. The spacing between the mirrors is adjusted such that it should be equal to the integral multiple of half wavelengths of the laser light.

(iii) Pumping system:

The pumping is done through electrical discharge by using electrodes that are connected to a high frequency alternating current source.

Energy Level Diagram of He-Ne Laser



Operation of He-Ne Laser

The common helium-neon gas laser achieves a population inversion in a different way. A mixture of about 10 parts of helium and 1 part of neon at a low pressure (1 torr) is placed in a glass tube that has parallel mirrors, one of them partly transparent, at both ends. The spacing of the mirrors is again (as in all lasers) equal to an integral number of half-wavelengths of the laser light. An electric discharge is produced in the gas by means of electrodes outside the tube connected to a source of high-frequency alternating current, and collisions with electrons from the discharge excite He and Ne atoms to metastable states respectively 20.61 and 20.66 eV above their ground states.

Operation of He-Ne Laser

Some of the excited He atoms transfer their energy to ground-state Ne atoms in collisions, with the 0.05 eV of additional energy being provided by the kinetic energy of the atoms. The purpose of the He atoms is thus to help achieve a population inversion in the Ne atoms.

The laser transition in Ne is from the metastable state at 20.66 eV to an excited state at 18.70 eV, with the emission of a 632.8-nm photon. Then another photon is spontaneously emitted in a transition to a lower metastable state; this transition yields only incoherent light.

Operation of He-Ne Laser

The remaining excitation energy is lost in collisions with the tube walls. Because the electron impacts that excite the He and Ne atoms occur all the time, unlike the pulsed excitation from the xenon flash lamp in a ruby laser, a He-Ne laser operates continuously.

Characteristics of He-Ne Laser

The He-Ne laser is a relatively low power device with an output in the visible red portion of the spectrum. The most common wavelength produced by He-Ne lasers is 632.8nm, although two lower power (1.152 μ m and 3.391 μ m) infrared wavelengths can be produced if desired. Majority of He-Ne lasers generate less than 10m watt of power, but some can be obtained commercially with up to 50m watts of power. For He-Ne lasers the typical laser tube is from 10 to 100 cm in length and the life time of such a tube can be as high as 20,000 hours.

Advantages of He-Ne Laser

- He-Ne laser has very good coherence property.
- He-Ne laser can produce three wavelengths that are $1.152\mu\text{m}$, $3.391\mu\text{m}$ and 632.8nm , in which the 632.8nm is most common because it is visible usually in red color.
- He-Ne laser tube has very small length approximately from 10 to 100cm and best life time of 20.000 hours.
- Cost of He-Ne laser is less from most of other lasers.
- Construction of He-Ne laser is also not very complex.
- He-Ne laser provide inherent safety due to low power output.

Disadvantages of He-Ne Laser

- It is relatively low power device means its output power is low.
- He-Ne laser is low gain system/ device.
- To obtain single wavelength laser light, the other two wavelengths of laser need suppression, which is done by many techniques and devices. So it requires extra technical skill and increases the cost also.
- High voltage requirement can be considered its disadvantage.
- Escaping of gas from laser plasma tube is also its disadvantage.

Applications of He-Ne Laser

The He-Ne laser is one of the most commonly used laser today because of the following applications.

- He-Ne lasers are produced in large quantities from many years.
- Many schools / colleges / universities use this type of laser in their science programs and experiments.
- He-Ne lasers also used in super market checkout counters to read bar codes and QR codes.
- The He-Ne lasers also used by newspapers for reproducing transmitted photographs.
- He-Ne lasers can be use as an alignment tool.
- It is also used in Guns for targeting.